**FYUGP B.Sc. MATHEMATICS HONOURS**

CONVEX OPTIMIZATION (MAT5EJ303 (2))

Semester V · Elective (Specialisation: Data Science)

**MODULE 1: REVIEW OF MULTIVARIABLE CALCULUS**

Past questions

## Section A: 1 to 3 marks

***Directional derivatives***

1 Find the directional derivative of $f(x,y) = x^2 \sin 2y$ at $\left(1, \frac{\pi}{2}\right)$ in the direction of $\vec{v} = 3\hat{i} - 4\hat{j}$. **(3 marks)** *March 2022*

***Gradient***

2 Find the gradient of $f(x,y) = y \ln x + xy^2$ at the point $(1,2)$. **(2 marks)** *March 2023*

3 Find the gradient of $f(x,y) = x^2 + y^2 + 1$ at the point $(1,2)$. Use the resutl to find the directional derivative of $f$ at $(1,2)$ in the direction from $(1,2)$ to $(2,3)$. [as printed] **(2 marks)** *March 2024*

***Jacobian***

4 Find the Jacobian for the change of variables defined by $x = r\cos\theta$, $y = r\sin\theta$. **(3 marks)** *March 2022*

***Stationary points and relative extrema***

5 Find the relative extrema of the function $f(x,y) = x^2 + y^2 - 2x + 4y$. **(3 marks)** *March 2022*

6 Find the relative extrema of $f(x,y) = 1 - \left(x^2 + y^2\right)^{1/3}$. **(2 marks)** *March 2023*

7 Find the critical points of $f(x,y) = -x^3 + 4xy - 2y^2 + 1$. **(2 marks)** *March 2024*

## Section B: 4 to 6 marks

***Chain rule***

8 Use the chain rule to find $\dfrac{dw}{dt}$, where $w = 2x^3y^2z$, $x = t$, $y = \cos t$, $z = t\sin t$. **(5 marks)** *March 2025*

***Stationary points and relative extrema***

9 Find the relative extrema of $f(x,y) = -x^3 + 4xy - 2y^2 + 1$. **(6 marks)** *March 2022*

10 Find the relative extrema of $f(x,y) = 2x^2 + y^2 + 8x - 6y + 20$. **(5 marks)** *March 2023*

11 Find the relative extrema of $f(x,y) = x^3 + y^2 - 2xy + 7x - 8y + 2$. **(5 marks)** *April 2026*